

Features

- Very high tracking accuracy
- Output voltage adjustable down to 2.0 V
- Stable with ceramic output capacitors
- Very low dropout voltage of typically 250 mV at 50 mA
- Very low current consumption of typically 3 μ A in standby mode
- Wide input voltage range $-16\text{ V} \leq V_{IN} \leq 45\text{ V}$
- Wide temperature range: $-40^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$
- Short circuit protected output (to GND and to battery)
- Reverse polarity protected input
- Overtemperature protection
- Green Product (RoHS compliant)

Description

The linear voltage regulator HE105 is a monolithic, integrated low-dropout voltage tracking regulator with high accuracy in small SOT23-6 package. The HE105 is designed to supply off-board systems, for example sensors in powertrain management systems under the severe conditions of automotive applications. The HE105 provides protection functions against reverse polarity as well as against short circuit to GND and to battery. The output voltage follows the reference voltage applied to the EN/ADJ input with very high accuracy, up to a supply voltage of 40 V and up to an output current of 50 mA. The required minimum reference voltage at EN/ADJ is 2.0 V .

Potential applications

- Automotive sensor supply
- Protected sensor supply for off-board sensors
- Secondary voltage supply for automotive ECU
- High precision voltage tracking
- Precision voltage replication
- Power switch for off-board load



Part number	Package	Marking	Packing
HE105RMAB	SOT23-6	HE105	3000/Tape&Reel

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1 Block diagram

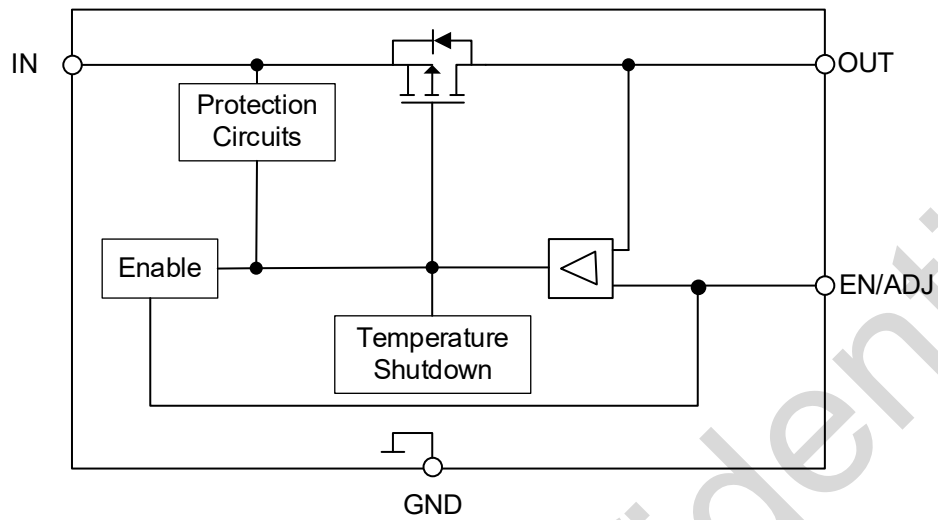


Figure 1-1 Block diagramPin configuration

1.1 Pin assignment

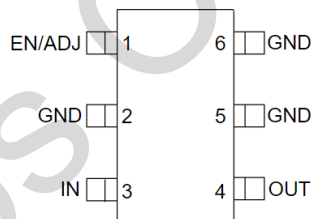


Figure 1-2 Pin configuration

1.2 Pin definitions and functions

Table 1-1 Pin definitions and functions

Pin	Symbol	Function
1	EN/ADJ	Enable/adjust: Connect the reference voltage to this pin. The reference voltage can be connected directly or via a voltage divider for lower output voltages. For the compensation of disturbances on the line, a capacitor close to the IC pins is recommended. "low" signal disables the device "high" signal enables the device
2	GND	Ground: Internally connected to pin5 and pin6. Connect to heatsink area.
3	IN	Input: Compensating line disturbances with a small ceramic capacitor to GND close to the IC terminals is recommended.
4	OUT	Tracker output 50 mA output current capability Connect to GND with a capacitor close to the pins, respecting capacitance and ESR requirements given in Table 2-2 Functional range.
5	GND	Ground: Internally connected to pin2 and pin6. Connect to heatsink area.
6	GND	Ground: Internally connected to pin2 and pin5. Connect to heatsink area.

2 General product characteristics

2.1 Absolute maximum ratings

Table 2-1 Absolute maximum ratings¹⁾

Symbol	Parameter	Note or condition	Values			Unit	Number
			Min.	Typ.	Max.		
Voltage							
V _{IN}	Input voltage	—	-16	—	45	V	P_3.1.1
V _{EN/ADJ}	Enable/Adjust voltage	—	-0.3	—	45	V	P_3.1.2
V _{OUT}	Output voltage	—	-5	—	45	V	P_3.1.3
V _{IN} -V _{OUT}	Input output voltage Difference	—	-30	—	45	V	P_3.1.4
Temperature							
T _j	Junction temperature	—	-40	—	150	°C	P_3.1.5
T _{stg}	Storage temperature	—	-55	—	150	°C	P_3.1.6
ESD							
V _{ESD,HBM}	ESD susceptibility	²⁾ Human Body Model (HBM)	-2	—	2	kV	P_3.1.7
V _{ESD,CDM}	ESD susceptibility	³⁾ CDM	-1	—	1	kV	P_3.1.8

T_J = -40°C to 150°C, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

¹⁾ Not subject to production test, specified by design.

²⁾ ESD susceptibility, HBM according to ANSI/ESDA/JEDEC JS001(1.5kΩ, 100pF)

³⁾ Charged device model (CDM) robustness according to ESDA STM5.3.1 or ANSI/ESD S.5.3.1

Note:

Stresses above the ones listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods of time may affect device reliability.

Integrated protection functions are designed to prevent IC destruction under fault conditions described in the datasheet. Fault conditions are considered as outside the normal operating range. Protection functions are not designed for continuous repetitive operation.

2.2 Functional range

Table 2-2 Functional range

Symbol	Parameter	Note or condition	Values			Unit	Number
			Min.	Typ.	Max.		
V_{IN}	Input voltage range	—	4	—	40	V	P_3.2.1
$V_{EN/ADJ}$	En/adjust input voltage range (Voltage tracking range)	—	2	—	20	V	P_3.2.2
C_{OUT}	Capacitance of output Capacitor	¹⁾	1	—	—	μF	P_3.2.3
ESR_{COUT}	Equivalent series resistance of output capacitor	²⁾	—	1	3	Ω	P_3.2.4
T_j	Junction temperature	—	-40	—	150	°C	P_3.2.5

¹⁾ The minimum output capacitance requirement is applicable for a worst case capacitance tolerance of 30%

²⁾ Relevant ESR value at $f = 10$ kHz.

Note: Within the functional range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given in the electrical characteristics table.

2.3 Thermal resistance

Table 2-3 Thermal resistance¹⁾

Symbol	Parameter	Note or condition	Values			Unit	Number
			Min.	Typ.	Max.		
R_{thJA}	Junction to ambient	²⁾	—	84	—	K/W	P_3.3.1
R_{thJA}	Junction to ambient	²⁾ Footprint only	—	228	—	K/W	P_3.3.2
R_{thJA}	Junction to ambient	300 mm ² PCB heatsink area	—	123	—	K/W	P_3.3.3
R_{thJA}	Junction to ambient	600 mm ² PCB heatsink area	—	109	—	K/W	P_3.3.4
R_{thJSP}	Junction to soldering point	Pins 2, 5 fixed to T_A	—	32	—	K/W	P_3.3.5

¹⁾ Not subject to production test, specified by design.

²⁾ Package mounted on PCB FR4; 80 × 80 × 1.5 mm³; 35 μm Cu, 5 μm Sn; horizontal position; zero airflow.

Note: This thermal data was generated in accordance with JEDEC JESD51 standards. For more information visit www.jedec.org.

2.4 Functional description tracking regulator

The regulator controls the output voltage V_{OUT} by comparing it to the reference voltage applied to the EN/ADJ pin and driving a pLDMOS pass transistor accordingly. The stability of the control loop depends on the following parameters:

- output capacitor C_{OUT}
- load current
- IC temperature
- poles and zeroes in the frequency response of the circuit including HE105
- external circuitry

An input capacitor C_{IN} is strongly recommended to buffer disturbances on the line.

To ensure stable operation, the output capacitor's capacitance and its equivalent series resistance ESR must fulfill the requirements in [Table 2-2](#). The output capacitor must be sized suitably to buffer load transients.

Connect each capacitor close to the pins.

The internal protection features are designed to protect the device itself as well as the application from destruction in case of catastrophic events. These safeguards contain the following:

- Output current limitation
- Reverse polarity protection
- Thermal shutdown

Output current limitation

In order to protect the pass element and the package from excessive power dissipation, the device limits the maximum output current at high input voltage.

Reverse polarity protection

The device allows a negative supply voltage.

Thermal shutdown

The overtemperature protection circuit is designed to prevent immediate destruction of the device in certain fault conditions (for example a permanent short circuit at output) by switching off the power stage. After the chip cools down, the regulator restarts. If the fault is not removed, it will lead to an oscillatory behavior of the output voltage. A junction temperature above 150°C is outside the maximum ratings and reduces the lifetime of the device.

3 Block description and electrical characteristics

3.1 Electrical characteristics tracking regulator

Table 3-1 Electrical characteristics tracking regulator

$V_{IN} = 13.5\text{ V}$, $2.0\text{ V} \leq V_{EN/ADJ} \leq 20\text{ V}$, $T_j = -40^\circ\text{C}$ to 150°C , all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Symbol	Parameter	Note or condition	Values			Unit	Number
			Min.	Typ.	Max.		
Tracking output							
ΔV_{OUT}	Output voltage tracking accuracy	$\Delta V_{OUT} = V_{EN/ADJ} - V_{OUT}$; $4\text{ V} \leq V_{IN} \leq 22\text{ V}$; $0.1\text{ mA} \leq I_{OUT} \leq 50\text{ mA}$; $2\text{ V} \leq V_{EN/ADJ} \leq V_{IN} - 0.5\text{ V}$; $V_{EN/ADJ} \leq 20\text{ V}$	-10	–	10	mV	P_4.1.1
ΔV_{OUT}	Output voltage tracking accuracy	$\Delta V_{OUT} = V_{EN/ADJ} - V_{OUT}$; $4\text{ V} \leq V_{IN} \leq 32\text{ V}$; $0.1\text{ mA} \leq I_{OUT} \leq 25\text{ mA}$; $2\text{ V} \leq V_{EN/ADJ} \leq V_{IN} - 0.5\text{ V}$; $V_{EN/ADJ} \leq 20\text{ V}$	-10	–	10	mV	P_4.1.2
ΔV_{OUT}	Output voltage tracking accuracy	$\Delta V_{OUT} = V_{EN/ADJ} - V_{OUT}$; $4\text{ V} \leq V_{IN} \leq 40\text{ V}$; $0.1\text{ mA} \leq I_{OUT} \leq 10\text{ mA}$; $2\text{ V} \leq V_{EN/ADJ} \leq V_{IN} - 0.5\text{ V}$; $V_{EN/ADJ} \leq 20\text{ V}$	-10	-	10	mV	P_4.1.3
$\Delta V_{OUT,load}$	Load regulation steady state	$I_{OUT} = 0.1\text{ mA}$ to 50 mA $V_{EN/ADJ} = 5\text{V}$	-3	–	–	mV	P_4.1.4
$\Delta V_{OUT,line}$	Line regulation steady state	$V_{IN} = 5.5\text{ V}$ to 32V ; $I_{OUT} = 10\text{mA}$; $V_{EN/ADJ} = 5\text{V}$	–	–	3	mV	P_4.1.5
PSRR	Power supply ripple rejection	¹⁾ $f_{\text{ripple}} = 100\text{ Hz}$; $V_{\text{ripple}} = 1\text{Vpp}$; $I_{OUT} = 10\text{mA}$; $C_{OUT} = 10\mu\text{F}$, ceramic type	–	85	–	dB	P_4.1.6
$I_{OUT,max}$	Output current limitation	$V_{OUT} = V_{EN/ADJ} - 0.1\text{V}$; $V_{EN/ADJ} = 5\text{V}$	51	85	120	mA	P_4.1.7
$I_{OUT,rev}$	Reverse current	$V_{IN} = 0\text{V}$; $V_{OUT} = 16\text{V}$; $V_{EN/ADJ} = 5\text{V}$	-100	-50	–	μA	P_4.1.10
$I_{IN,rev}$	Reverse current at negative input voltage	$V_{IN} = -16\text{V}$; $V_{OUT} = 0\text{V}$; $V_{EN/ADJ} = 5\text{V}$	-10	0	–	μA	P_4.1.11
V_{dr}	Dropout voltage	$V_{dr} = V_{IN} - V_{OUT}$; $I_{OUT} = 50\text{mA}$; $V_{EN/ADJ} = 5\text{V}$	–	250	450	mV	P_4.1.12
Overtemperature protection							
$T_{j,sd}$	Overtemperature shutdown threshold		–	175	–	$^{\circ}\text{C}$	P_4.1.13
$T_{j,sdh}$	Overtemperature shutdown threshold		–	15	–	$^{\circ}\text{C}$	P_4.1.14

¹⁾ Not subject to production test, specified by design

²⁾ Measured when the output voltage V_O has dropped 100 mV from the nominal value obtained at $V_I = 13.5\text{ V}$.

3.2 Typical performance characteristics tracking regulator

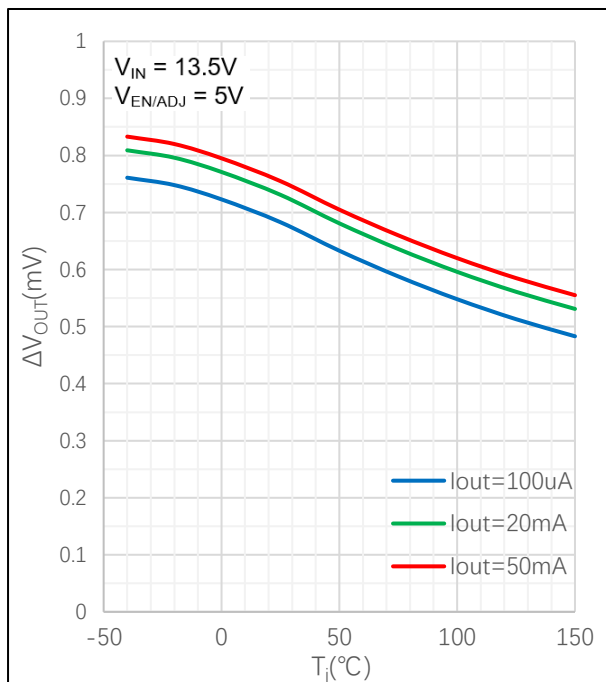


Figure 3-1 Tracking accuracy ΔV_{OUT} versus junction temperature T_j

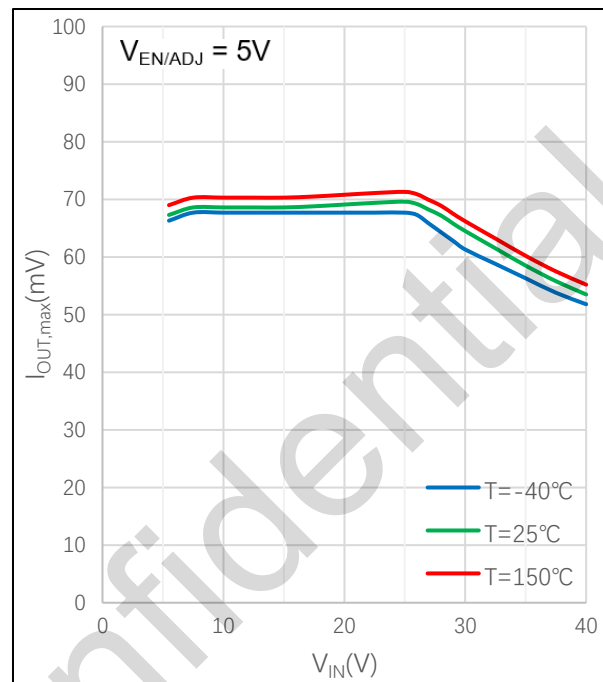


Figure 3-2 Output current limitation $I_{OUT,max}$ versus input voltage V_{IN}

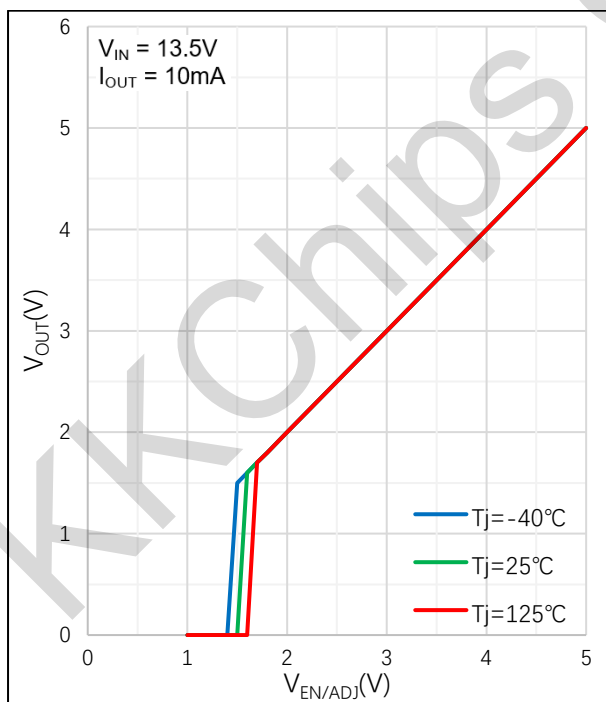


Figure 3-3 Output voltage V_{OUT} versus enable/adjust voltage $V_{EN/ADJ}$

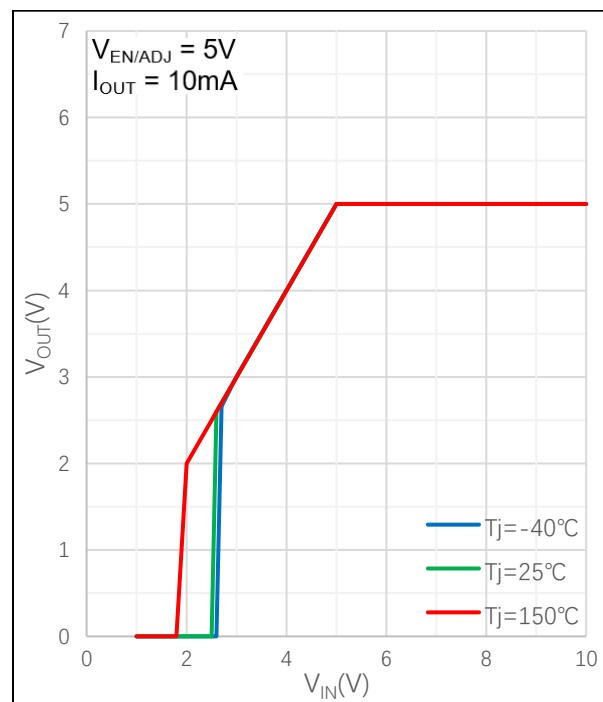


Figure 3-4 Output voltage V_{OUT} versus input voltage V_{IN}

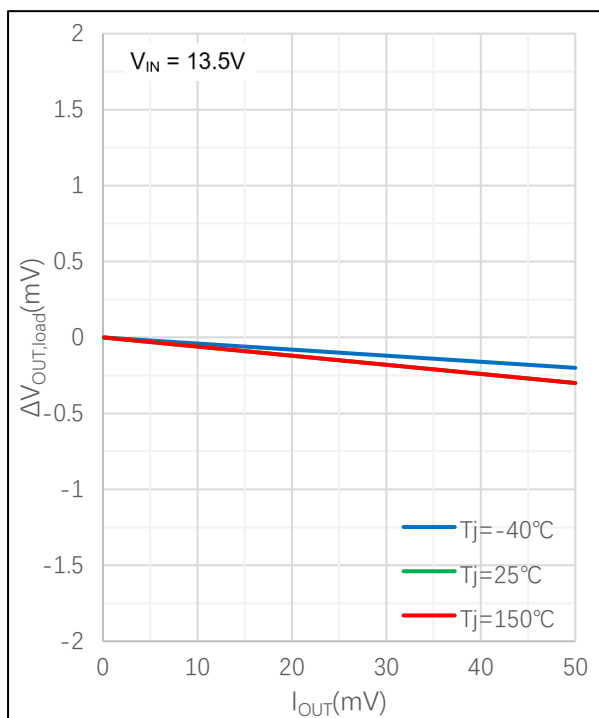


Figure 3-5 Load regulation $\Delta V_{OUT,load}$ versus output current I_{OUT}

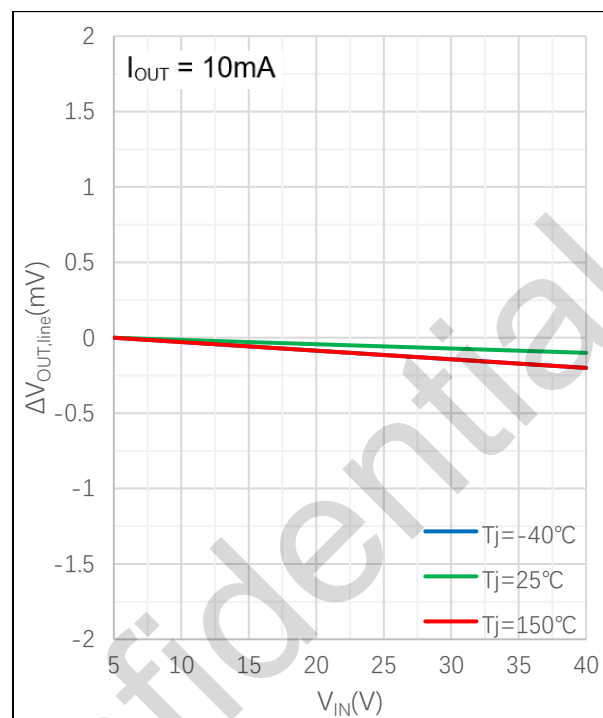


Figure 3-6 Line regulation $\Delta V_{OUT,line}$ versus input voltage V_{IN}

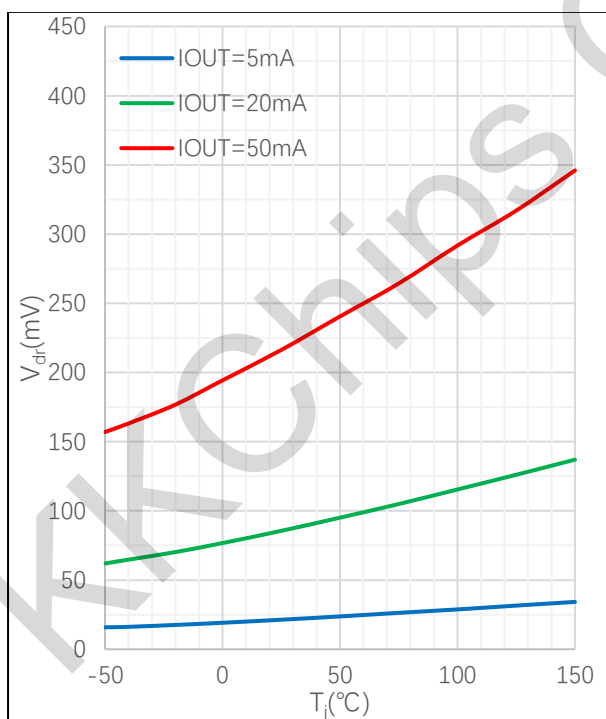


Figure 3-7 Dropout voltage V_{dr} versus junction temperature T_j

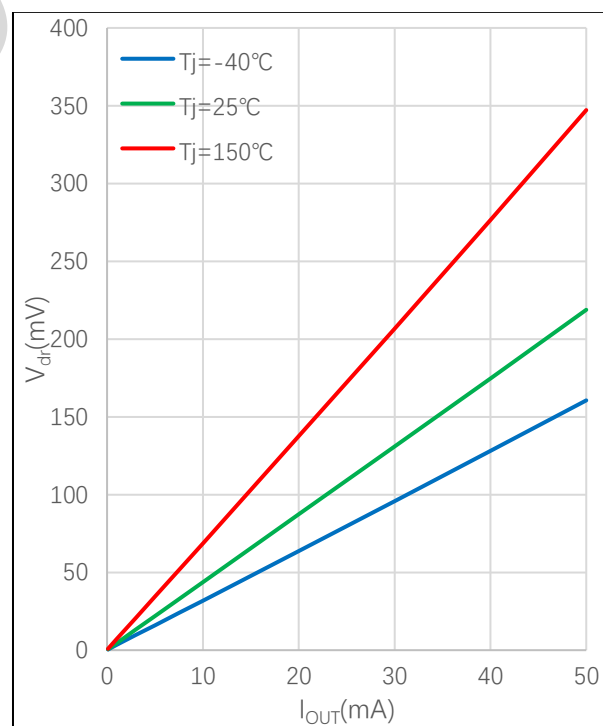


Figure 3-8 Dropout voltage V_{dr} versus output current I_{OUT}

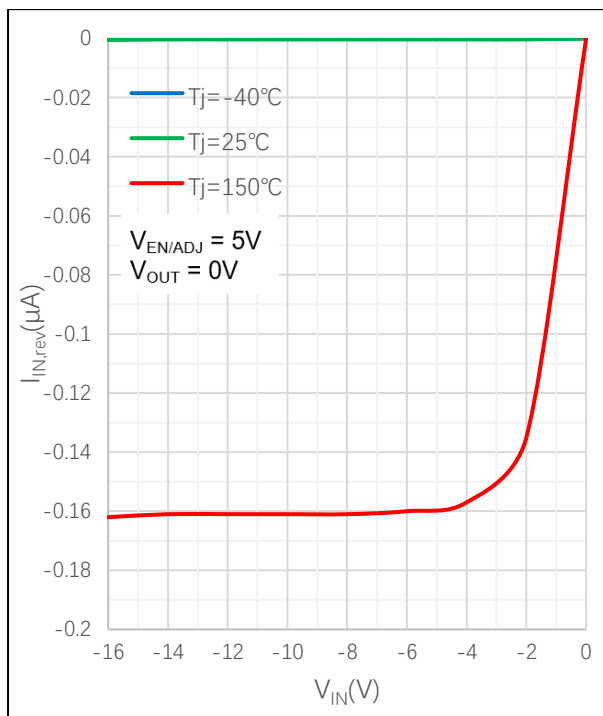


Figure 3-9 Reverse input current $I_{IN,rev}$ versus input voltage V_{IN}

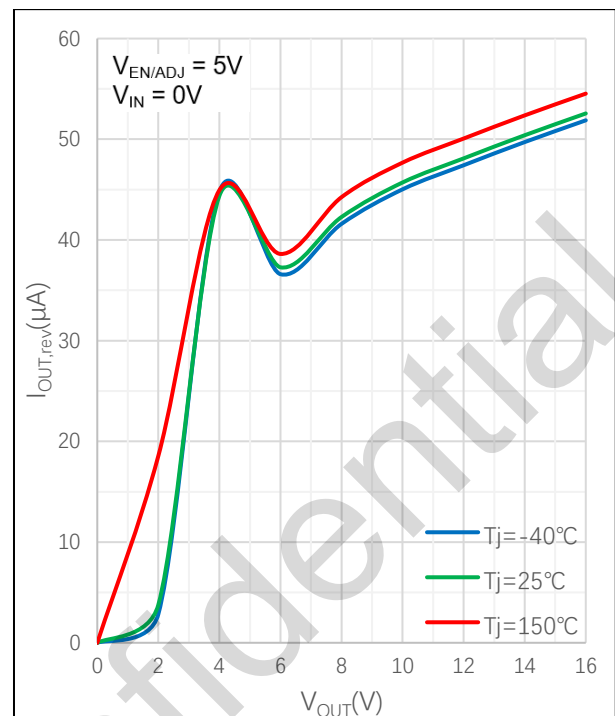


Figure 3-10 Reverse output current $I_{OUT,rev}$ versus output voltage V_{OUT}

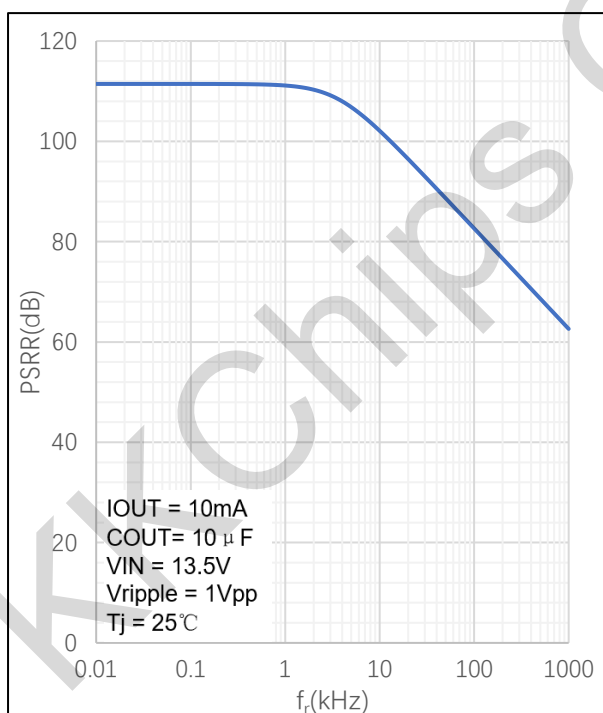


Figure 3-11 Power supply ripple rejection PSRR versus ripple frequency f_r

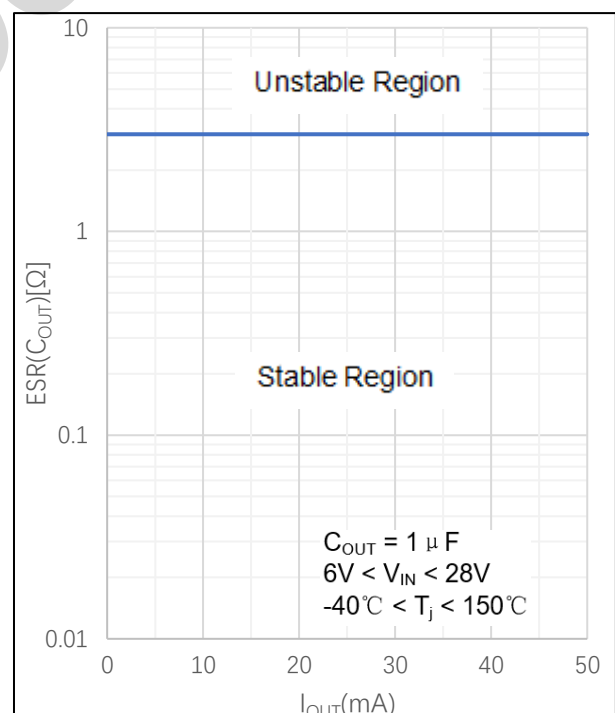


Figure 3-12 Output capacitor equivalent series resistance ESR_{COUT} versus output current I_{OUT}

3.3 Electrical characteristics tracking regulator

Table 3-2 Electrical characteristics current consumption

$V_{IN} = 13.5\text{ V}$, $2.0\text{ V} \leq V_{EN/ADJ} \leq 20\text{ V}$, $T_j = -40^\circ\text{C}$ to 150°C , all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Symbol	Parameter	Note or condition	Values			Unit	Number
			Min.	Typ.	Max.		
$I_{q,off}$	Current consumption standby mode	$I_{q,off} = I_{IN}$; $V_{EN/ADJ} \leq 0.4\text{ V}$; $T_j \leq 125^\circ\text{C}$	—	0.3	1.5	μA	P_4.2.1
I_q	Current consumption	$I_q = I_{IN} - I_{OUT}$; $I_{OUT} \leq 0.1\text{ mA}$; $V_{EN/ADJ} = 5\text{ V}$; $T_j \leq 125^\circ\text{C}$	—	40	75	μA	P_4.2.2
I_q	Current consumption	$I_q = I_{IN} - I_{OUT}$; $I_{OUT} \leq 50\text{ mA}$; $V_{EN/ADJ} = 5\text{ V}$	—	288	500	μA	P_4.2.3

3.4 Typical performance characteristics current consumption

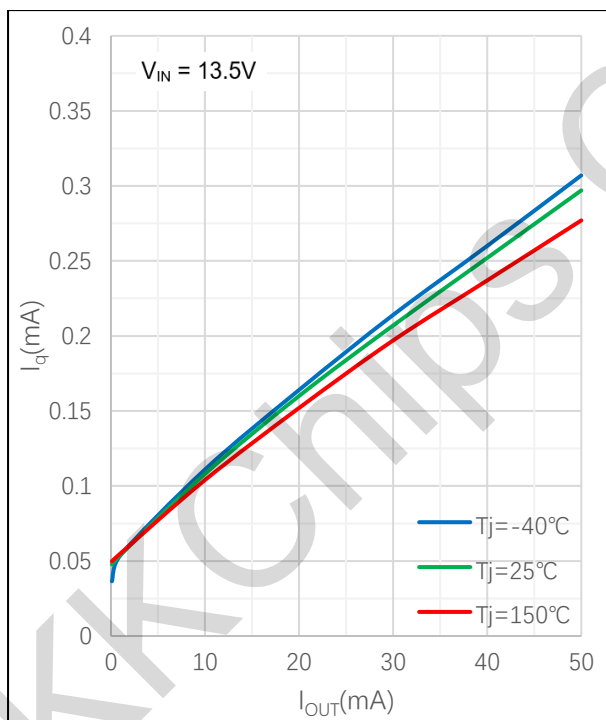


Figure 3-13 Current consumption I_q versus output current I_{OUT}

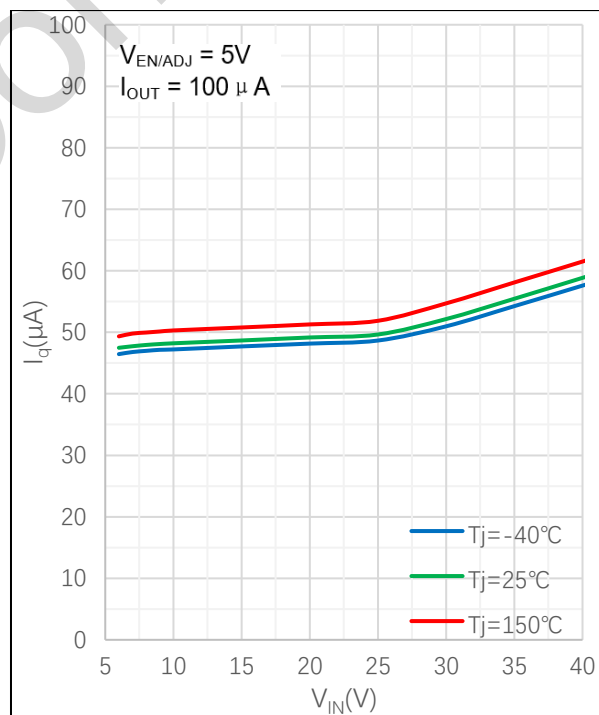


Figure 3-14 Current consumption I_q versus input voltage V_{IN}

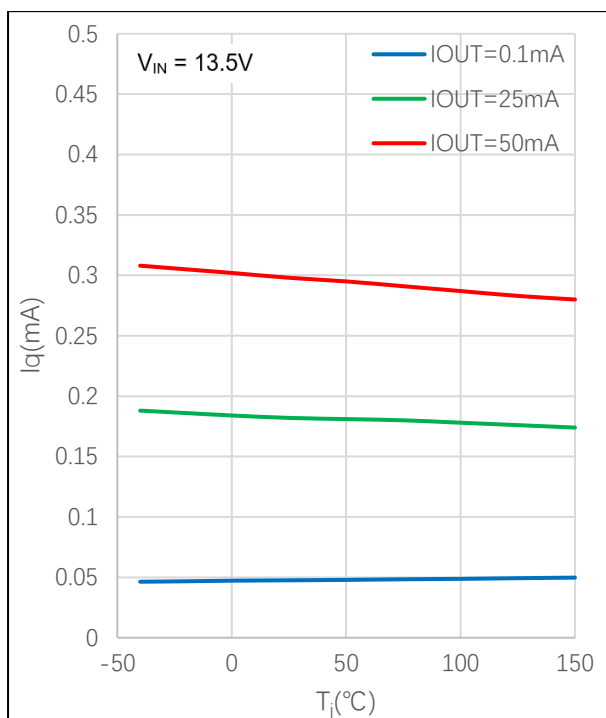


Figure 3-15 Current consumption I_q versus junction temperature T_j

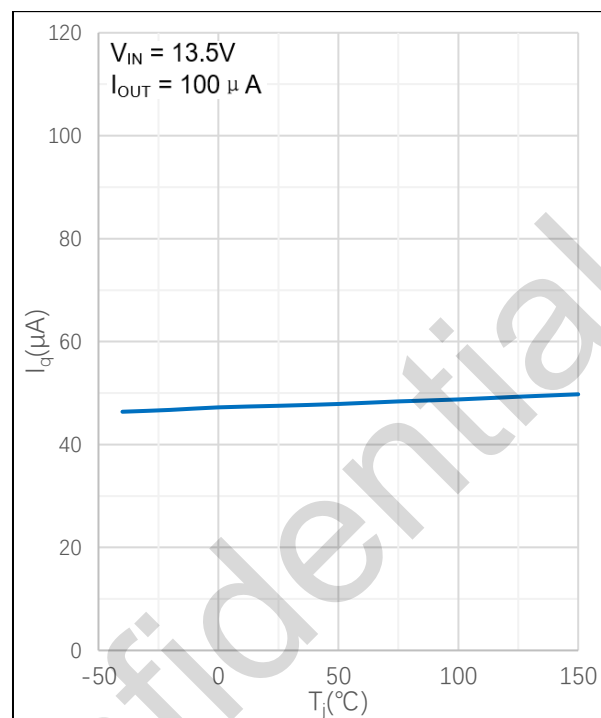


Figure 3-17 Current consumption I_q versus junction temperature T_j (I_{OUT} low)

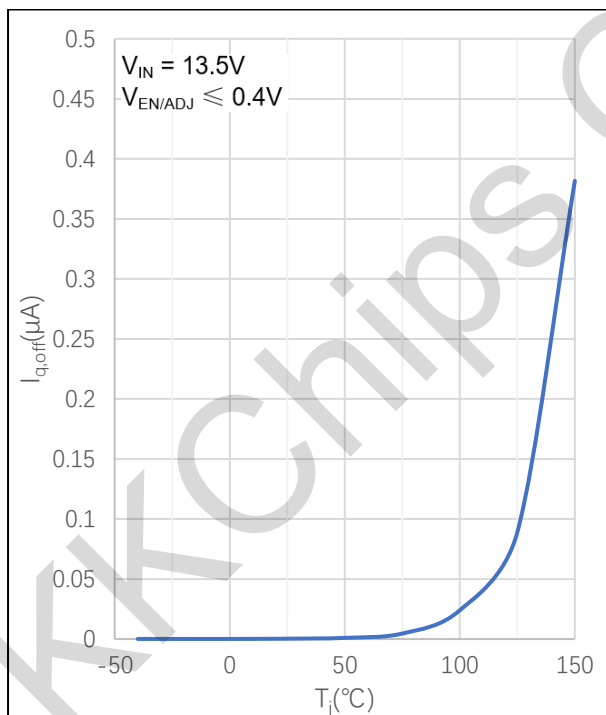


Figure 3-16 Current consumption in OFF mode $I_{q,off}$ versus junction temperature T_j

3.5 Functional description enable/adjust input

On a “low” signal at the enable/adjust input EN/ADJ the device switches to standby mode in order to minimize the quiescent current.

If the EN/ADJ pin is not connected, then the “low” level from the internal pull-down resistor switches off the regulator.

3.6 Electrical characteristics enable/adjust input

Table 3-3 Electrical characteristics enable/adjust input

$V_{IN} = 13.5\text{ V}$, $2.0\text{ V} \leq V_{EN/ADJ} \leq 20\text{ V}$, $T_J = -40^\circ\text{C}$ to 150°C , all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Symbol	Parameter	Note or condition	Values			Unit	Number
			Min.	Typ.	Max.		
$V_{EN/ADJ,off}$	Enable/adjust off voltage range	$V_{OUT} = 0\text{V}$	–	–	0.8	V	P_4.3.1
$V_{EN/ADJ,on}$	Enable/adjust on voltage range	V_{OUT} settled	2	–	–	V	P_4.3.2
$I_{EN/ADJ}$	Current consumption	$V_{EN/ADJ} = 5\text{ V}$	–	3	5	μA	P_4.3.3

3.7 Typical performance characteristics enable/adjust input

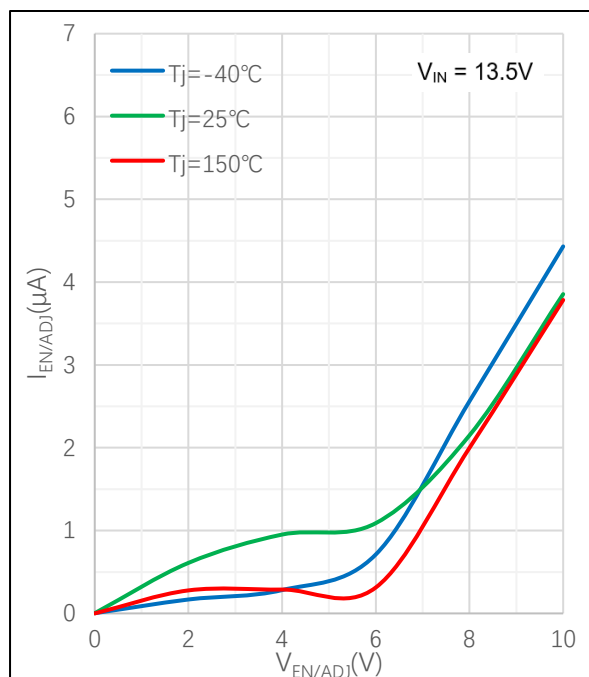


Figure 3-18 Enable/adjust input current $I_{EN/ADJ}$ versus voltage $V_{EN/ADJ}$

4 Application information

Note: The following information is given as an example for the implementation of the device only and shall not be regarded as a description or warranty of a certain functionality, condition or quality of the device.

4.1 Application diagram

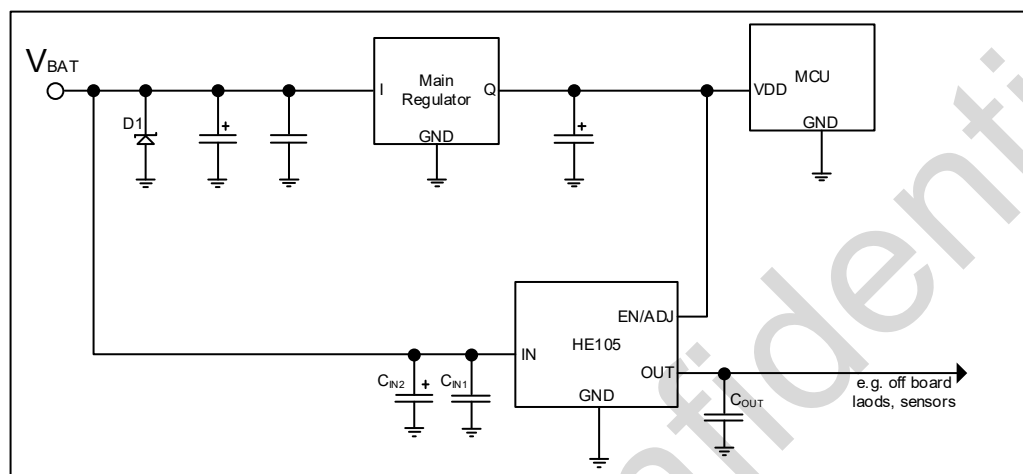


Figure 4-1 Application circuit

Note: This figure is a simplified example of an application circuit. The function must be verified in the application.

4.2 Selection of external components

4.2.1 Input pin

Figure 4-1 shows a typical input circuitry for a voltage tracking regulator. The following external components at the input are recommended in case of possible external disturbance.

Ceramic capacitor

A ceramic capacitor C_{IN1} (100 nF to 470 nF) at the input filters high frequency disturbance imposed by the line, such as ISO pulses 3a/b. Place C_{IN1} as close as possible to the input pin of the voltage tracking regulator on the PCB.

Aluminum electrolytic capacitor

An aluminum electrolytic capacitor C_{IN2} (10 μ F to 470 μ F) at the input smoothens high energy pulses, such as ISO pulse 2a. Place C_{IN2} close to the input pin of the voltage tracking regulator on the PCB.

Overvoltage suppression diode

A suitably sized diode $D1$ suppresses high voltage beyond the maximum ratings of the circuit components and protects the device from damage due to overvoltage.

4.2.2 Output pin

An output capacitor C_{OUT} is necessary for the stability of the voltage tracking regulator, see [Table 2-2](#). The typical performance graph [Output capacitor equivalent series resistance ESRCOUT versus output current IOUT](#) shows the stable operating range of the device.

In an automotive environment, ceramic capacitors with X5R or X7R dielectrics are recommended.

Place C_{OUT} on the same side of the PCB as the device and as close as possible to both the OUT pin and the GND pin.

In case of rapid transients of the input voltage or of the load current, C_{OUT} must be dimensioned properly to ensure the output stability in the application.

4.2.3 Enable/Adjust pin

[Figure 4-1](#) shows the typical input circuitry for a voltage tracking regulator. Typically the enable/adjust pin is connected to a fixed voltage reference that the regulator tracks. In the example of the application diagram EN/ADJs is connected to the supply voltage of a microcontroller. Alternatively, the voltage reference can also be adjusted by a voltage divider.

4.3 Thermal considerations

Knowing the input voltage, the output voltage and the load profile of the application, the total power dissipation can be calculated by:

$$P_D = (V_{IN} - V_{OUT}) \times I_{OUT} + V_{IN} \times I_q$$

Equation 1

with

P_D : continuous power dissipation

V_{IN} : input voltage

V_{OUT} : output voltage

I_{OUT} : output current

I_q : quiescent current

The maximum acceptable thermal resistance R_{thJA} can then be calculated by:

$$R_{thJA,max} = \frac{T_{j,max} - T_a}{P_D}$$

Equation 2

with

$T_{j,max}$: maximum allowed junction temperature

T_a : ambient temperature

Based on the above calculation the proper PCB type and the necessary heat sink area can be determined with reference to the specification in [Table 2-3](#).

Example

Application conditions:

$$V_{IN} = 13.5 \text{ V}$$

$$V_{OUT} = V_{EN/ADJ} = 5 \text{ V}$$

$$I_{OUT} = 20 \text{ mA}$$

$$T_a = 125^\circ\text{C}$$

Calculation of $R_{thJA,max}$:

$$P_D = (V_{IN} - V_{OUT}) \times I_{OUT} + V_{IN} \times I_q = (13.5\text{V} - 5\text{V}) \times 20\text{mA} + 13.5\text{V} \times 0.2\text{mA} = 0.1727\text{W}$$

Equation 3

$$R_{thJA,max} = \frac{T_{j,max} - T_a}{P_D} = \frac{150^\circ\text{C} - 125^\circ\text{C}}{0.1727\text{W}} = 144.76\text{K/W}$$

Equation 4

As a result, the PCB design must ensure a thermal resistance $R_{thJA,max}$ lower than 144.76 K/W.

According to [Table 2-3](#) at least 300 mm² heatsink area is required on the FR4 1s0p PCB, or the FR4 2s2p board can be used.

5 Package information

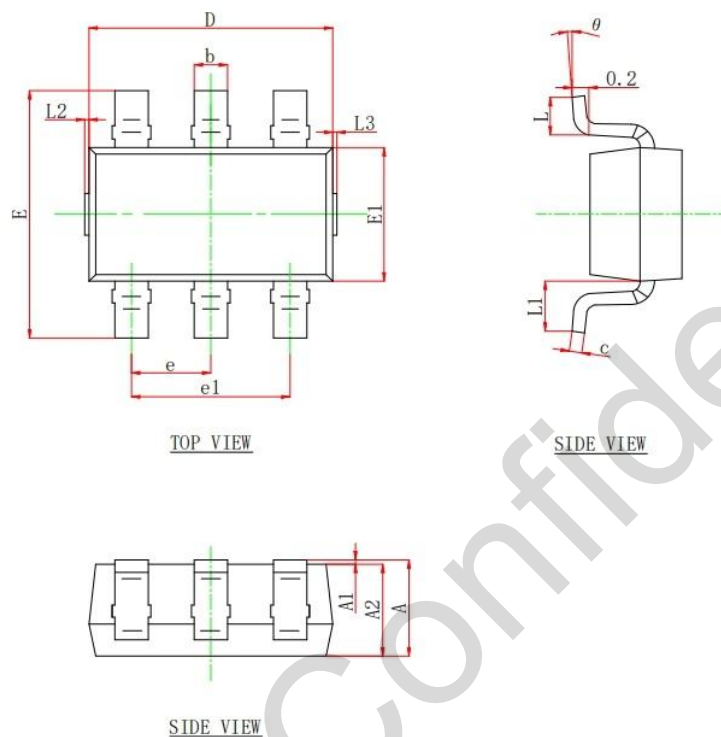


Figure 5-1 SOT23-6

Table 5-1 SOT23-6 mechanical

Symbol	Min.(mm)	Max.(mm)
A	1.050	1.250
A1	0.000	0.100
A2	1.050	1.150
b	0.300	0.500
c	0.100	0.200
D	2.820	3.020
E1	1.500	1.700
E	2.650	2.950
e	0.950(BSC)	
e1	1.800	2.000
L	0.300	0.600
L1	0.600REF	
θ	0°	8°

Green Product (RoHS compliant)

To meet the world-wide customer requirements for environmentally friendly products and to be compliant with government regulations the device is available as a Green Product. Green Products are RoHS compliant (Pb-free finish on leads and suitable for Pb-free soldering according to IPC/JEDEC J-STD-020).

6 Recommended pad size

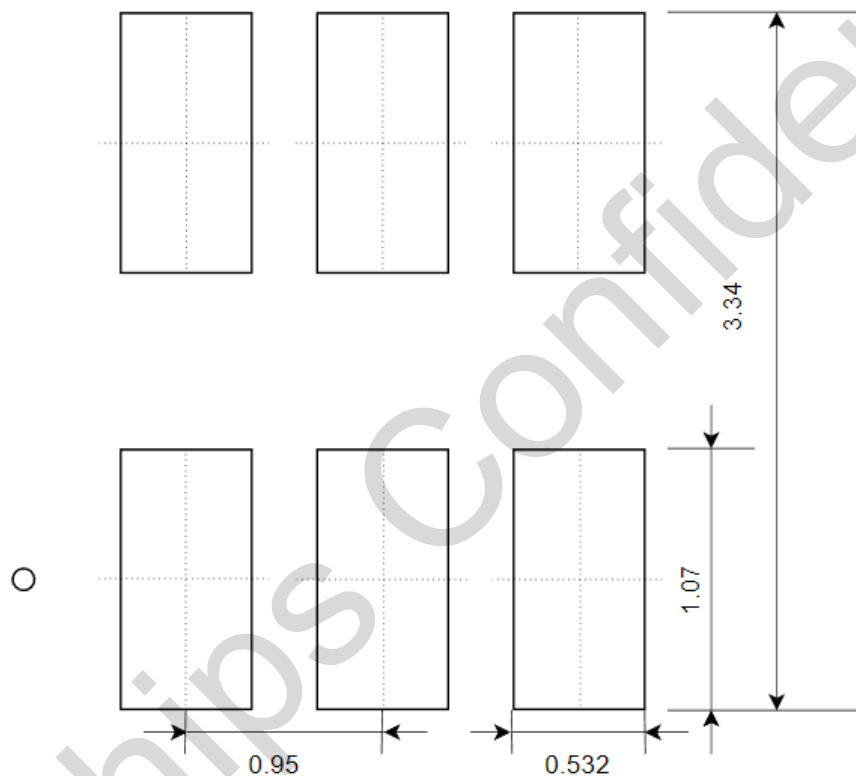


Figure 6-1 Recommended pad size

7 Customer service

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8 Revision history

Revision	Date	Changes
1.0	2025/5/16	• Change to the official version 1.0.

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